

ME5412 Robotics for Healthcare

Module briefing

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Module Syllabus and Schedule

Week	Date	CA	Topic	Lecturer
Week 1	11/8/2026		Course briefing, introduction to healthcare robotics, principles of Assistive Technology	Cecilia Laschi
Week 2	18/8/2026		Interfaces for assistive devices, mobility aids	Cecilia Laschi
Week 3	25/8/2026		Manipulation aids, upper limb prostheses	Cecilia Laschi
Week 4	1/9/2026		Introduction to stroke and stroke rehabilitation	Haoyong Yu
Week 5	7/9/2026		Lower limb rehabilitation robotics	Haoyong Yu
Week 6	15/9/2026		Upper limb rehabilitation robotics	Haoyong Yu
Recess				
Week 7	29/9/2026		Project assignment	Both
Week 8	6/10/2026	50%	Mid-Term Quiz (1h30)	Both
Week 9	13/10/2026		Project time and consultation	Both (on demand)
Week 10	21/10/2026		Project time and consultation	Both (on demand)
Week 11	27/10/2026		Project time and consultation	Both (on demand)
Week 12	3/11/2026	50%	Project presentations	Both
Week 13	10/11/2026		Project presentations cntd.	Both

Quiz (assessed, 50%)

- Online quiz on **ExamSoft***
- In person, open book, 1h30 duration, October 6th, 2026
- Questions on the topics of the lectures, open book

*Students' **Briefing Sessions:** <https://nus.atlassian.net/wiki/x/O4BXAQ>

Self-learning: briefing slides or [Exemplify Student Briefing Video](#)

Student guides:

- Digital Assessment Student
Wiki: <https://nus.atlassian.net/wiki/spaces/DAsstudent/overview>
- Exemplify Student Briefing Slides: <https://nus.atlassian.net/wiki/x/M4BXAQ>
(Refer to In-Person Assessment)
- Download and Install Exemplify: <https://nus.atlassian.net/wiki/x/HYBXAQ>
- How do I login to Exemplify? <https://nus.atlassian.net/wiki/x/7YBXAQ>
- Common Briefing Sessions: <https://nus.atlassian.net/wiki/x/O4BXAQ>
- Practice Exams: <https://nus.atlassian.net/wiki/x/IIBXAQ>

Project Assignment (assessed, 50%)

- Groups* of 5 students
- Each group free to choose a topic, within the course scope
- Prototypes not expected, but concept/design/simulation
- Presentation by each group, to the other students**, in 10-15 min
- Submission of slides as a report, by each group, November 3rd, 2026

*Please organize yourselves in groups and start setting up the groups in Canvas → ME5412 → People → Groups → ME5412 Project.

** All students are supposed to attend all presentations.

ME5412 Robotics for Healthcare

Introduction to Healthcare Robotics

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Robots for Healthcare

为何需要

- Why healthcare robotics, why now
- What counts as a "healthcare robot" — and the autonomy spectrum 什么样新医疗机器人
- A taxonomy: five families of healthcare robots 五大类别
- Core technical challenges across all five 共同技术挑战.
- Regulatory and ethical landscape 法规, 伦理.
- Course roadmap why → what → Type → Challenges

Why Healthcare Robotics, Why Now

- Three converging pressures are pulling robotics out of the lab and into the clinic



人口老龄化

2.1B

people worldwide will be aged 60+ by 2050 (WHO)



医疗工作者短缺

10M

projected global health-worker shortfall by 2030 (WHO)



护士大量非临床工作。

~35%

of a nurse's shift spent on non-clinical tasks that robots can offload

- Robotics won't replace clinicians — it extends scarce clinical time, reduces physical strain, and enables care that scales with demand.

机器人不会取代临床医护人员；机器人是拓展本就稀缺的医护人力，减轻人员身体负担，让医疗服务可以匹配不断增长的需求。

A Growing Healthcare Robotics Market (2025-2035)

医疗机器人市场持续增长（预测周期：2025-2035）

• 复合年增长率：14.6% • 2025年市场规模：165.5亿美元 • 2035年预测市场规模：652.9亿美元

FORECAST PERIOD

2025-2035

CAGR

14.6%

两大驱动因素：

1. 机器智能技术越来越多地融入外科手术流程
2. 医保报销制度改革，让资金投入大的机器人设备，中型医疗机构也有能力采购使用。

2025 MARKET SIZE

USD 16.55 Billion (2025)

2035 MARKET SIZE

USD 65.29 Billion (2035)

Two converging forces:

1. accelerating integration of machine intelligence into surgical workflows
 2. wave of reimbursement reforms that are making capital-intensive robotic platforms financially accessible to mid-tier healthcare facilities.
- 美国医疗保险和医疗补助服务中心（CMS）在2024年扩充了机器人辅助手术的医保报销编码。
 - The U.S. Centers for Medicare & Medicaid Services (CMS) expanded coverage codes for robot-assisted procedures in 2024,
 - 欧盟委员会在“地平线欧洲”计划下拨款12亿欧元，用于下一代临床自动化技术研发。
 - The European Commission allocated EUR 1.2 billion under Horizon Europe for next-generation clinical automation.
 - 北美医疗机器人市场预计占据38.1%营收份额：成熟的医保报销体系、教学医院数量众多。
 - North America Medical Robotics Market expected to hold 38.1% revenue share, due to well-established reimbursement routes and a high concentration of teaching hospitals.
 - Asia-Pacific is the fastest-expanding market with a CAGR of 16.1%. Government-supported rehabilitation initiatives in China, India and Japan drive growth.
 - 亚太市场增长速度最快，复合年增长率16.1%；中国、印度、日本政府大力扶持康复相关技术，拉动市场发展。
 - Europe ranks second with a 27.5% share, bolstered by unified MDR regulatory clearances and cross-border surgical training consortia.
 - 欧洲市场份额27.5%，位列第二；得益于统一的医疗器械法规（MDR）审批机制，以及跨国外科手术培训联盟。
 - The ability of manufacturers to implement predictive analytics into platform ecosystems will define the next decade.
 - 未来十年，厂商能否把预测分析能力整合进机器人平台生态，将成为行业发展的关键。

什么才算"医疗机器人"? ——工作定义

What Counts as a "Healthcare Robot"?

WORKING DEFINITION

A sensing–reasoning–acting system that autonomously or semi-autonomously performs physical tasks within clinical, rehabilitative, or care-related settings, in direct or indirect interaction with patients, clinicians, or care environments.

- 感知 (Sense) —— 探测患者、工具或环境信息;
 - **Sense** Perceives patients, tools, or environment
- 推理 (Reason) —— 在不确定性与安全限制下规划行动;
 - **Reason** Plans actions under uncertainty & safety limits
- 执行 (Act) —— 物理操作、移动或辅助。
 - **Act** Physically manipulates, moves, or assists

工作定义：一套具备感知—推理—执行能力的系统，可以自主或者半自主地在临床、康复或护理相关场景中执行物理任务。

SPECTRUM OF AUTONOMY 自主能力分级谱系

Tele-operated

Human drives every motion (e.g. remote surgery console)

1. 遥操作：人控制全部动作（例如远程手术控制台）

Supervised / shared control

Robot assists, filters tremor, enforces safety zones

2. 监督/共享控制：机器人辅助，过滤抖动，设置安全区域

Task autonomy

Robot executes a defined task, human sets goals

3. 任务自主：机器人完成指定任务，由人设定目标

Full autonomy

Robot perceives, decides, and acts with no human in the loop

4. 完全自主：机器人感知、决策、执行，全程无需人为介入

A Taxonomy of Healthcare Robots

01	02	03	04	05
Surgical & Interventional	Rehabilitation & Assistive	Hospital Service & Logistics	Socially Assistive & Companion	Diagnostic & Monitoring
Precision manipulation inside the body	Restoring or augmenting movement	Delivery, disinfection, transport	Cognitive, emotional, social support	Sensing, imaging, remote exam
1. 手术与介入机器人：人体内部高精度操作	2. 康复与辅助机器人：恢复、增强运动能力	3. 医院服务与物流机器人：物资配送、消毒转运	4. 社交辅助陪伴机器人：认知、情绪、社交支持	5. 诊断监测机器人：传感成像、远程检查

Surgical & Interventional Robots

消除手部抖动，实现高精度操作，拓展外科医生的操作范围，实现微创手术

Extending the surgeon's hands with tremor-free precision and minimally invasive access

1

Teleoperated systems

Surgeon-controlled manipulators (e.g. da Vinci) scale motion and filter tremor for minimally invasive surgery

1. 遥操作系统：医生操控机械臂（如达芬奇），缩放动作并过滤抖动，用于微创手术

2

Steerable & flexible instruments

Catheters, endoscopes, and continuum robots that navigate through natural or small-incision pathways

2. 可转向柔性器械：导管、内窥镜、连续体机器人，经由人体自然腔道或小切口进入体内

3

Orthopedic guidance robots

Semi-autonomous cutting/drilling within a surgeon-defined and imaging-verified boundary

3. 骨科引导机器人：在医生划定、影像确认的边界内半自动完成切割钻孔

4

Autonomous sub-tasks

Suturing or tissue-tracking research systems performing narrow steps under supervision

4. 自主子任务：研究类系统，在监督下完成缝合、组织追踪等特定操作

达芬奇 SP 机器人

WATCH [What is the da Vinci SP robotic system?](https://www.youtube.com/watch?v=TGjnb86HndU) — Intuitive Surgical, YouTube <https://www.youtube.com/watch?v=TGjnb86HndU>

康复与辅助机器人

Rehabilitation & Assistive Robots

恢复身体机能、提升行动能力，帮助人们实现独立生活

Restoring function, augmenting mobility, and supporting independent living

Robotic exoskeletons

Wearable powered limbs for gait training and daily mobility after stroke or spinal cord injury

Upper-limb rehab robots

End-effector and exoskeleton devices that guide repetitive, task-specific arm therapy

Smart prosthetics

Myoelectric or neurally controlled limbs with adaptive grip and terrain sensing

Assistive mobility & manipulation

Robotic wheelchairs and feeding/manipulation arms for people with limited motor function

- 机器人外骨骼：可穿戴动力肢体，用于中风、脊髓损伤后的步态训练和日常行走
- 上肢康复机器人：末端执行器、外骨骼设备，引导手臂完成重复性针对性康复训练
- 智能假肢：肌电或者神经信号控制假肢，自适应抓取，具备地形感知
- 辅助移动与操作设备：机器人轮椅、喂食操作机械臂，服务运动功能受限人群

外骨骼

WATCH Find Your Strength // Ekso Bionics — Ekso Bionics, YouTube <https://www.youtube.com/watch?v=RtBaQEKcguk>

医院服务与物流机器人

Hospital Service & Logistics Robots

The unglamorous work — moved, cleaned, delivered — that frees clinical staff 承担搬运、清洁、配送等基础工作，解放临床医护人员

Autonomous mobile robots (AMRs)

Navigate hospital corridors to move supplies, meds, linens, and lab samples

UV / disinfection robots

Autonomously sterilize rooms with UV-C light between patients

Delivery & pharmacy robots

Move medication trays and equipment between departments on demand

Telepresence units

Mobile screen-and-camera platforms for remote rounding and consults

- 自主移动机器人AMR：在医院走廊自主导航，运送物资、药品、布草、化验样本

- 紫外线消毒机器人：病人出院后，使用紫外C光自动对病房灭菌

- 药房配送机器人：按需跨科室运送药盘与医疗设备

- 远程会诊终端：带屏幕摄像头的移动设备，实现远程查房会诊

WATCH [Meet Moxi the Robot from Diligent Robotics](https://www.youtube.com/watch?v=TASOLpNR-Gw) — Diligent Robotics, YouTube <https://www.youtube.com/watch?v=TASOLpNR-Gw>

社交辅助与陪伴机器人

Socially Assistive & Companion Robots

Robots whose primary function is social, emotional, or cognitive — not physical

Eldercare companions

Reminders, check-ins, and social engagement for older adults living alone

Autism & pediatric therapy robots

Structured, predictable social partners for skill-building interventions

Dementia & cognitive support

Prompting, routine support, and calming interaction for cognitive decline

Coaching & mental-health support

Conversational agents for engagement in therapy or wellness programs

核心作用为社交、情绪、认知支持，而非物理作业

- 老年陪伴机器人：提醒事项、日常问候，陪伴独居老人社交互动
- 自闭症与儿童治疗机器人：行为模式稳定可控，作为社交伙伴帮助儿童训练社交技能
- 痴呆认知辅助机器人：提醒日常流程，安抚认知衰退患者情绪
- 心理辅导机器人：对话交互，辅助心理治疗与健康管理

■ **WATCH** [This robotic therapy seal is revolutionizing elderly care](https://www.youtube.com/watch?v=mj_-8yFxxA)

— Reuters, YouTube https://www.youtube.com/watch?v=mj_-8yFxxA

核心技术挑战 Core Technical Challenges

Every category above shares the same open problems

以上五类机器人共同存在待解决难题

Safety under uncertainty

Acting near fragile, unpredictable humans with hard real-time guarantees

不确定环境下的安全：在脆弱、行为不可预测的人类周边作业，需要硬实时安全保障

Human-robot interaction

Trust, transparency, and shared control between clinician/patient and machine

人机交互：医生、患者与机器之间建立信任、行为透明、实现共享控制

Perception in the clinic

Sensing tissue, people, and cluttered environments reliably and safely

临床环境感知：在杂乱环境中可靠、安全识别人体组织、人员物体

Regulatory approval

Certifying a learning, physically acting system as a medical device

监管审批：将具备学习能力、可进行物理动作的系统认证成为医疗器械

Data privacy & security

Protecting sensitive health data collected and acted on by autonomous systems

数据隐私安全：保护自主机器人采集、处理的敏感医疗健康数据

Cost & accessibility

Moving beyond flagship hospitals to equitable, affordable deployment

成本与普及性：走出大型标杆医院，做到价格亲民、广泛部署

监管与伦理现状

Regulatory & Ethical Landscape

A healthcare robot is a medical device, a co-worker, and a data system all at once

一台医疗机器人，同时是医疗器械、工作协作者、数据系统。

监管层面 REGULATORY

Device classification

FDA/EMA pathways (e.g. 510(k), De Novo) for autonomy levels that didn't exist when the rules were written

Liability 责任划分：机器人辅助治疗出现事故，责任归厂商、医院还是医生？
When a robot-assisted outcome goes wrong: manufacturer, hospital, or clinician?

Informed consent

Patients need to understand what the robot does — and what it doesn't
知情同意：患者必须清楚机器人能做什么，不能做什么

设备分类：FDA、欧盟医疗器械局审批路径（510(k)、De Novo）；现有法规制定时尚未出现各类机器人自主等级

ETHICAL 伦理层面

Autonomy vs. oversight

How much decision-making should shift from clinician to system, and when?

Equity of access

Will advanced care robotics widen or narrow gaps between well- and under-resourced settings?

Dignity & deception

Especially for companion robots — does simulated empathy mislead vulnerable users?

机器自主和人为监管：多大程度的决策可以交给机器，哪些情况不行？

医疗公平：高端医疗机器人会拉大还是缩小贫富地区医疗差距？

人格尊严与误导风险：尤其陪伴机器人，模拟共情是否会误导弱势群体用户？

Course Scope

01

Surgical & Interventional

Precision
manipulation inside
the body

02

Rehabilitation & Assistive

Restoring or
augmenting
movement

03

Hospital Service & Logistics

Delivery,
disinfection,
transport

04

Socially Assistive & Companion

Cognitive,
emotional, social
support

05

Diagnostic & Monitoring

Sensing, imaging,
remote exam